Tools and APIs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

TEACHER GUIDE · 1 OF 9

Tools and APIs

> Prepare Tools and APIs: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of tools and **api contract**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Conversation State and Memory

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conversation state and retention

Reference example, not the only acceptable wording: Teacher explanation: Conversation state stores facts or prior interactions for later use.

It needs scope, provenance, correction and retention rules rather than indefinite accumulation. This classroom assistant keeps an approved room ID only for the current session.

A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of building a grounded answer

Reference example, not the only acceptable wording: Teacher explanation: A grounded answer connects each factual claim to **evidence** actually retrieved for the task.

Retrieval success does not automatically make generation faithful. Build a claim-**evidence** table before final wording.

Preserve source versions and qualifiers, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conflicting and time-sensitive sources

Reference example, not the only acceptable wording: Teacher explanation: Changing facts require a stated time and source-version policy. Current-looking records can conflict without enough **evidence** to choose between them.

For this toy task a signed amendment supersedes the referenced original from its effective time. Keep both records. A newer timestamp alone has no authority unless the policy grants it.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

Concept to explain

A **tool call** is a structured request to an external capability with returned data.

Term	Meaning
API	An agreed interface through which software requests data or actions from another system.
tool call	A structured request to a tool with specified arguments.
argument	A value supplied to a function or tool to tell it what to operate on.
error response	A message or structured result indicating that a request could not be completed as expected.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain tools and apis using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Tools and APIs using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Tool schema requires operation and two numbers.

2. Request sends operation=divide, a=18, b=24.

3. Tool returns result=0.75; product formats 75%.

4. Timeout or invalid **input** must trigger retry, repair, or safe explanation—not invented data.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

The teaching case

Choose and verify tools.

Exact explanation to model

Why the result follows

Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Explain the added worked example

Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
API contract	Declared input, output and failure rules for an interface
Tool call	A proposed or executed operation with structured arguments
Not found	An explicit absence result rather than a fabricated value

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Proposed call lookup_room has room_id=B and extra **argument** book=true.

Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Tools and APIs

Check a library lookup call before it reaches a tool.

The facts we will use

Supplied facts

lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/21

Tools and APIs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check shape

Exactly book_id is supplied, and C7 is text.



2. Check membership

C7 occurs in the authorised registry.



3. Read only

Run the lookup. This tool contract contains no lending or deletion operation.

CHANGE ONE THING

Add lend: true to the same call.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check shape	Exactly book_id is supplied, and C7 is text.
2. Check membership	C7 occurs in the authorised registry.
3. Read only	Run the lookup. This tool contract contains no lending or deletion operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A valid **tool call** can still return unavailable or outdated records. Validate the response and access rights separately.

Change one thing in the pictured case

Change one thing

Change: Add lend: true to the same call.

Predicted result

Prediction: Reject the call before invoking the tool.

Why the result follows

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Tools and APIs, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

2. State the decision to explain

Write the question to explain: Choose and verify tools.

3. Reveal the worked reasoning

Reveal after **prediction**s: Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Hold back the independent task

Keep for an independent attempt: A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

> result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage	Teacher action
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. ID validity and operation permission are separate checks; reject the extra field under this contract. Changed-case check: Reject the call before invoking the tool.

An extra operation is outside the exact allowed contract, even when book_id is valid.

Changed-case reasoning: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Tools and APIs investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Match requests to schemas, validate **argument**s, simulate successful and failed calls, and design a maximum retry rule.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Tools and APIs questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Independent question

> **Independent question · withhold the key**
>
> A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> A read-only availability lookup is authorized. Placing an order is a different action and requires explicit authorization.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Access to a tool automatically authorizes every action it supports.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Calculate 102 and verify 17×6. Retrieve the relevant roots passage and cite its supported answer. Check tool **input**s and returned records. Neither a calculator nor a search tool should be given unrelated action permissions.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts.

W2 Trace the worked case

Workbook question · W2

Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

Teacher reasoning · W4

Teacher explanation: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect and extend learning

> Connect Tools and APIs to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Allow only retrieval and calculator tools with validated **argument**s and visible results.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 22: Requests, Responses, and JSON. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: First fails type; second is valid **input** returning **not found**; third fails extra-field validation. **Not found** is not capacity zero. Require intermediate reasoning and the stated scope.

Fresh retry: Room M exists with measured capacity zero; room N is absent. Distinguish results. Expected reasoning: M found with capacity 0; N **not found** with no measured capacity. Preserve zero separately from missing.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Conversation state and retention. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/22, ai-3/24, ai-3/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Tools and APIs lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of tools and **api contract**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Proposed call lookup_room has room_id=B and extra **argument** book=true. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

Check every learner before discussion: Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Add lend: true to the same call. Require a **prediction** and a reason before comparing. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

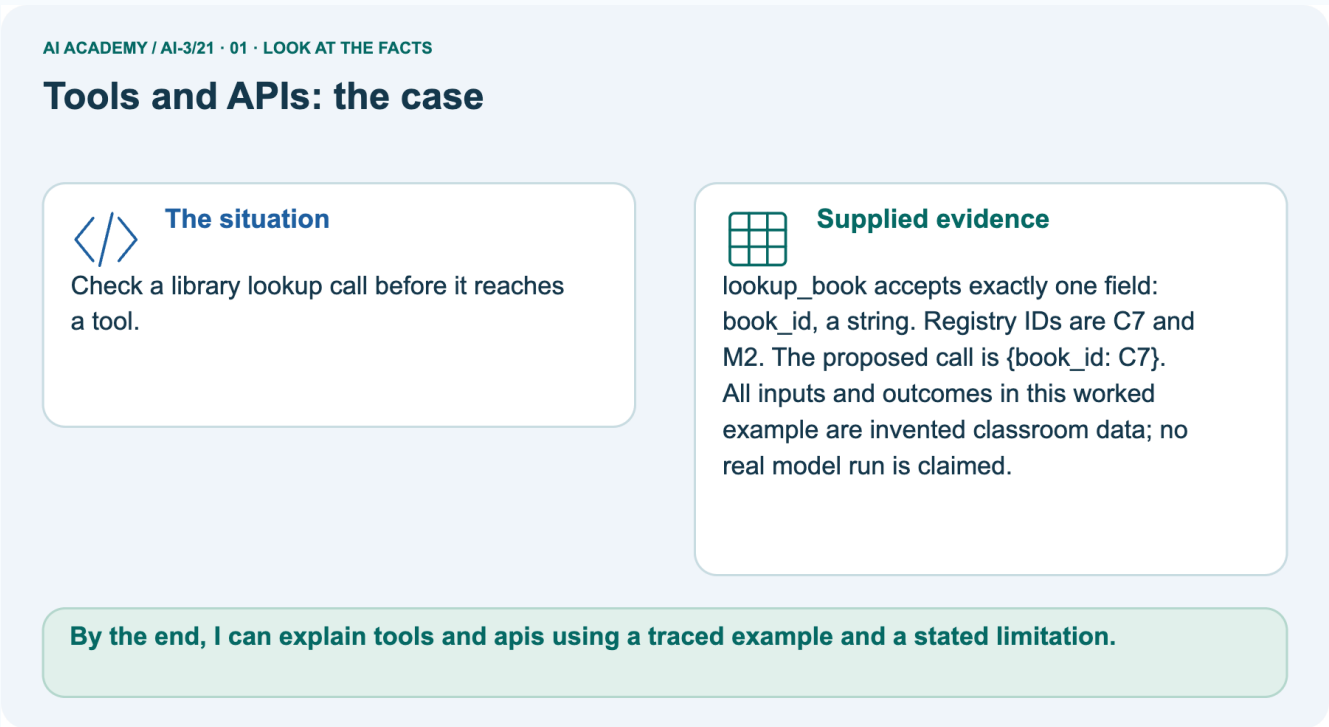
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Tools and APIs graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Tools and APIs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Check a library lookup call before it reaches a tool. Supplied facts: lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

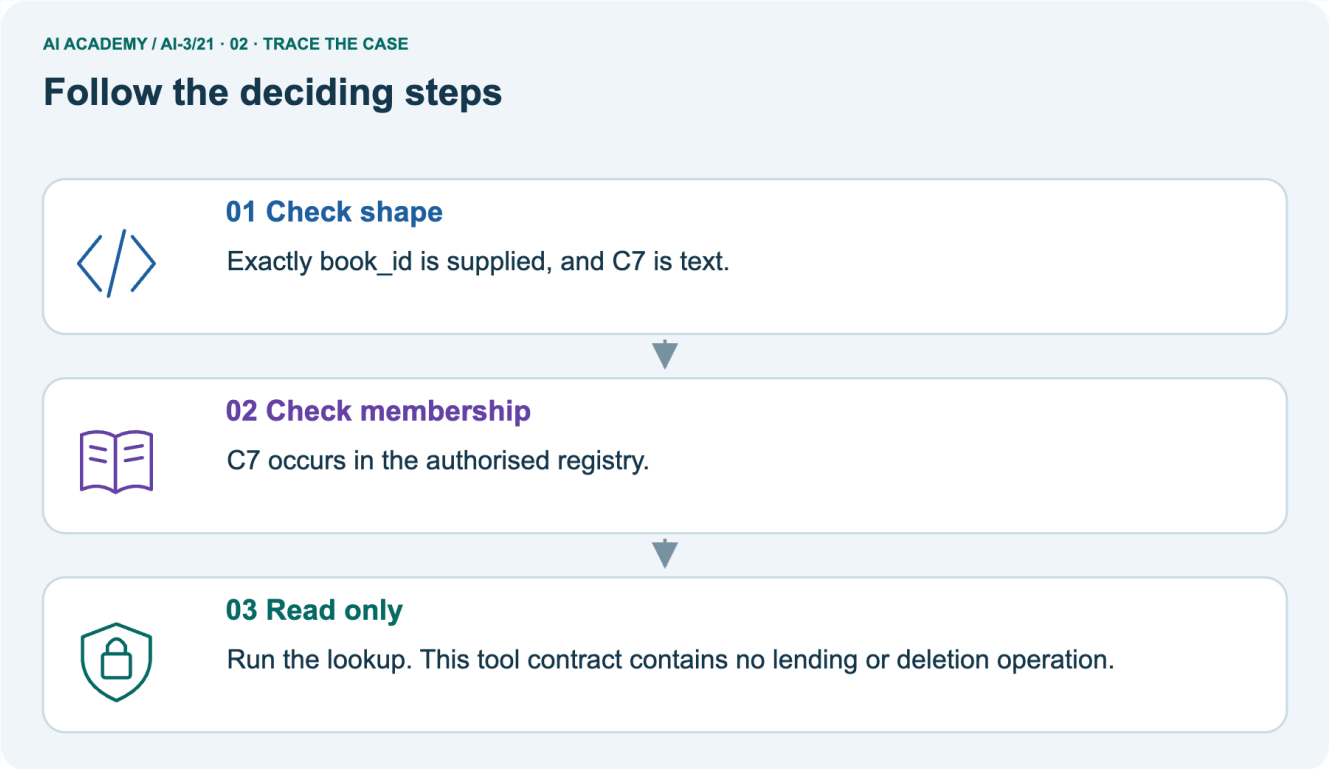
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

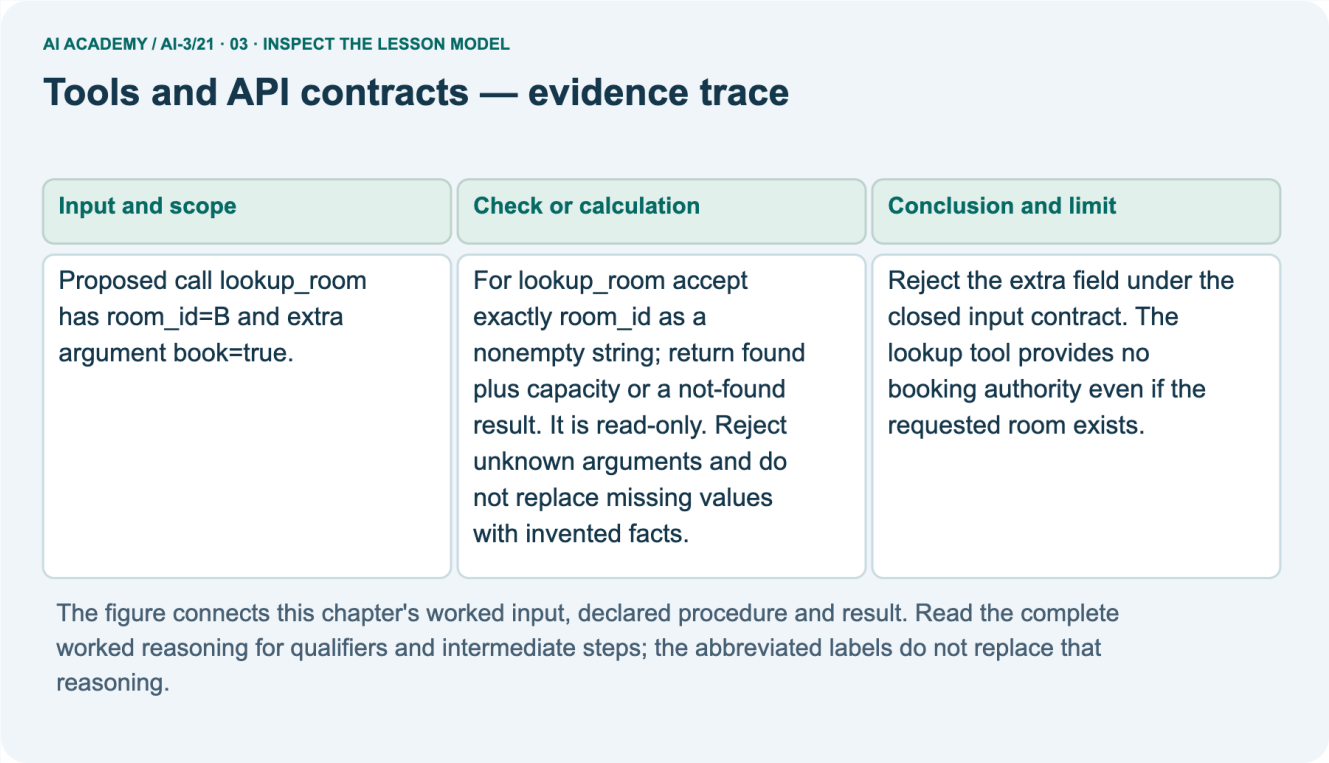
Check shape: Exactly book_id is supplied, and C7 is text. → Check membership: C7 occurs in the authorised registry. → Read only: Run the lookup. This tool contract contains no lending or deletion operation.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Tools and API contracts — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Tools and **API contract**s — **evidence** trace. Proposed call lookup_room has room_id=B and extra **argument** book=true. | For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result.

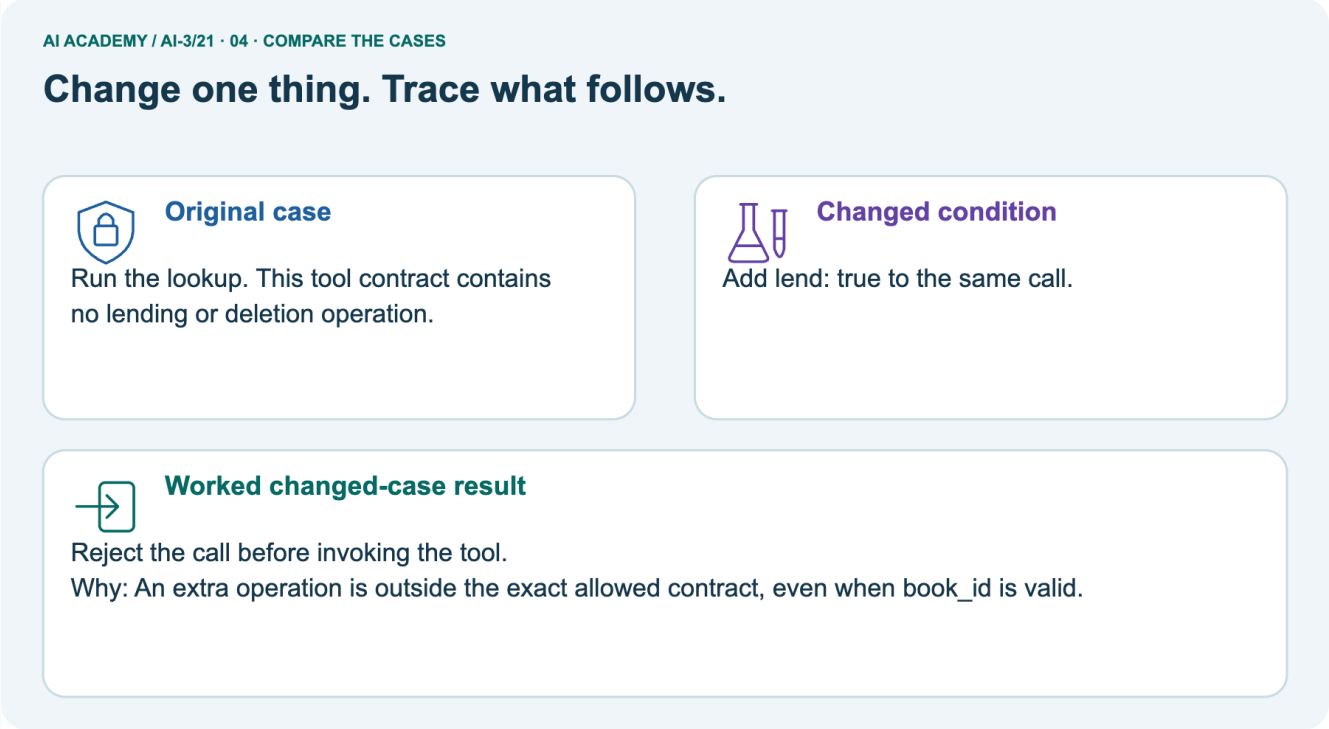
It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts. | Reject the extra field under the closed **input** contract.

The lookup tool provides no booking authority even if the requested room exists.

> Read the labels and values. Explain the relationship the model shows.

> **Discuss while showing the picture**
>
> Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Run the lookup. This tool contract contains no lending or deletion operation. Changed condition: Add lend: true to the same call. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

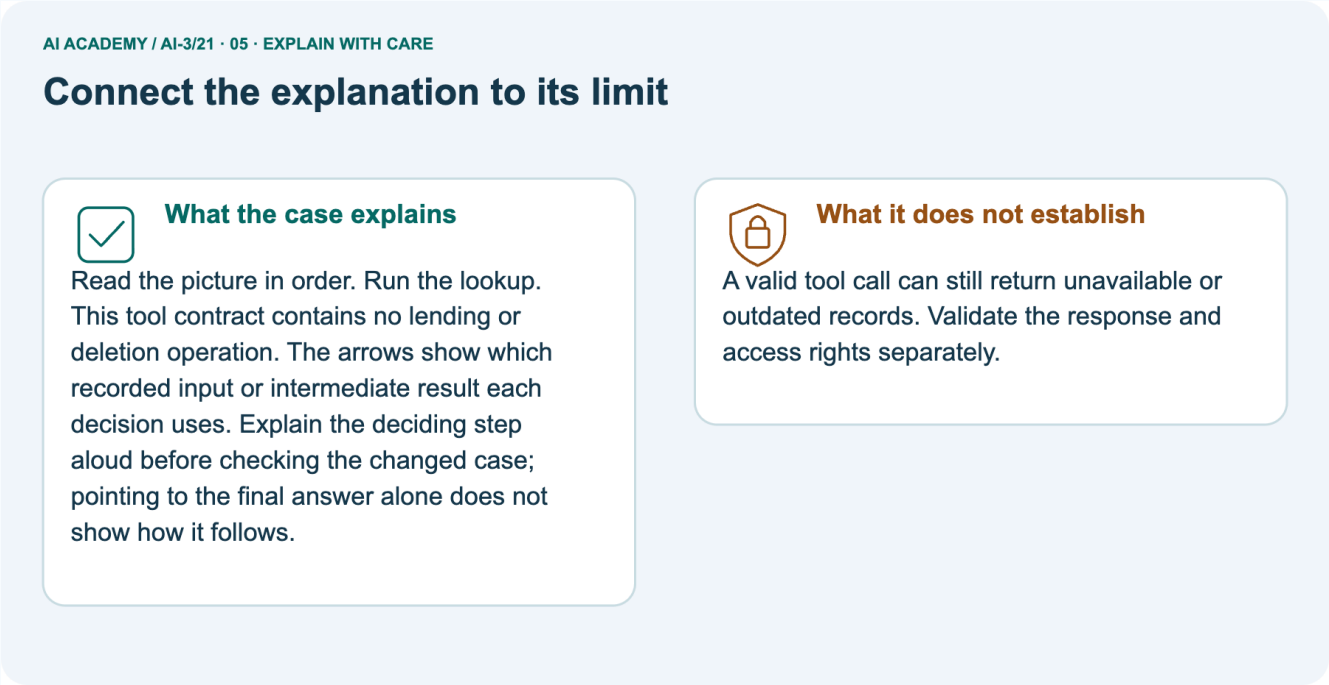
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: An extra operation is outside the exact allowed contract, even when book_id is valid.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A valid **tool call** can still return unavailable or outdated records.

Validate the response and access rights separately.

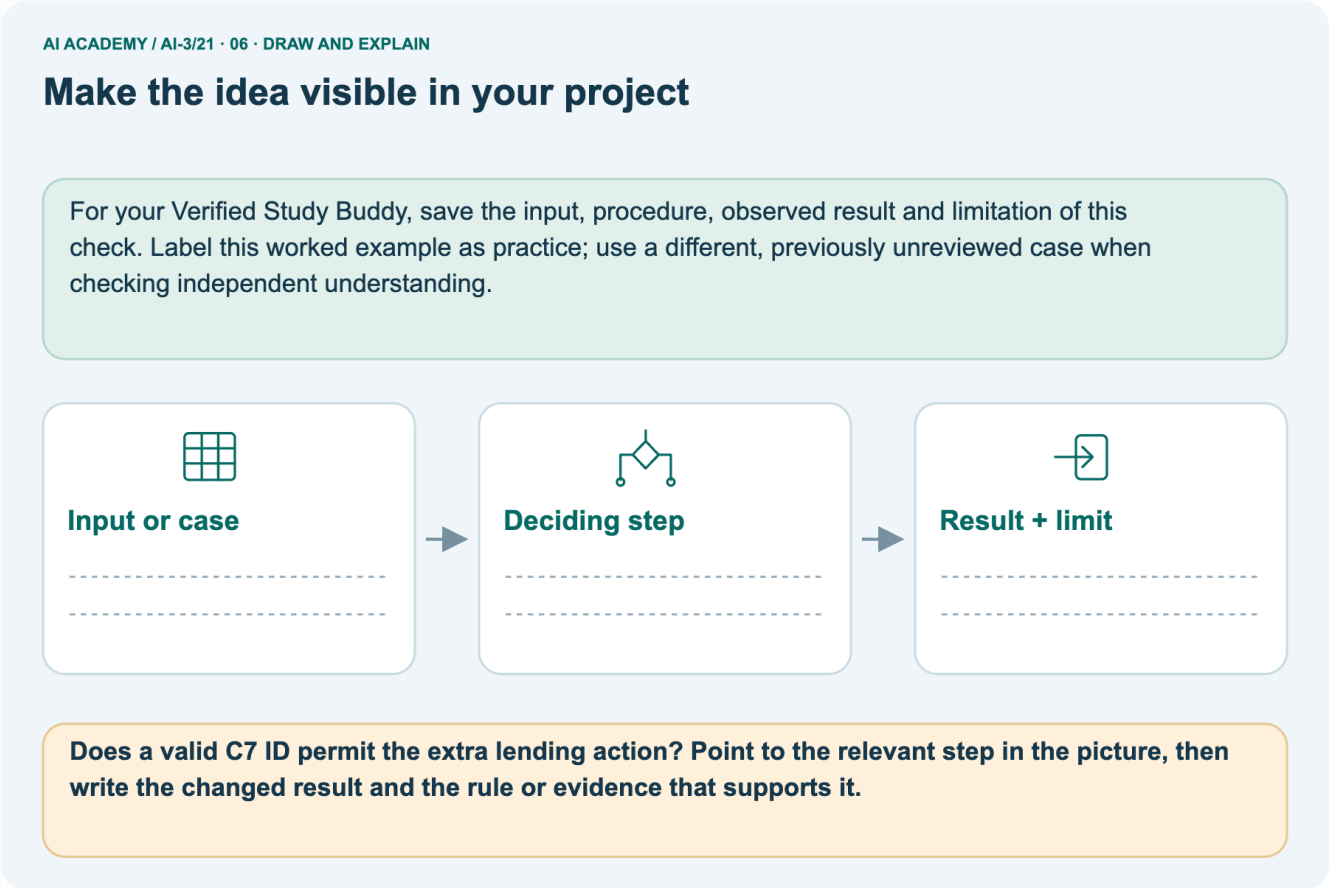
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: A valid **tool call** can still return unavailable or outdated records. Validate the response and access rights separately.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. ID validity and operation permission are separate checks; reject the extra field under this contract. Changed-case check: Reject the call before invoking the tool.

An extra operation is outside the exact allowed contract, even when book_id is valid.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.